

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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1 Introduction

Difference-in-Differences (DiD) has become one of the most popular research designs used to evaluate causal effects of policy interventions. In its canonical format, there are two time periods and two groups: in the first period no one is treated, and in the second period some units are treated (the treated group), and some units are not (the comparison group). If, in the absence of treatment, the average outcomes for treated and comparison groups would have followed parallel paths over time (which is the so-called parallel trends assumption), one can estimate the average treatment effect for the treated subpopulation (ATT) by comparing the average change in outcomes experienced by the treated group to the average change in outcomes experienced by the comparison group. Methodological extensions of DiD methods often focus on this standard two periods, two groups setup; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), [Athey and Imbens \(2006\)](#), [Qin and Zhang \(2008\)](#), [Bonhomme and Sauder \(2011\)](#), [de Chaisemartin and D’Haultfœuille \(2017\)](#), [Botosaru and Gutierrez \(2018\)](#), [Callaway et al. \(2018\)](#), and [Sant’Anna and Zhao \(2020\)](#).¹

Many DiD empirical applications, however, deviate from the canonical DiD setup and have more than two time periods and variation in treatment timing. In this article, we provide a unified framework for average treatment effects in DiD setups with multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. We concentrate our attention on DiD with staggered adoption, i.e., to DiD setups such that once units are treated, they remain treated in the following periods.

The core of our proposal relies on separating the DiD analysis into three separate steps: (i) identification of policy-relevant disaggregated causal parameters; (ii) aggregation of these parameters to form summary measures of the causal effects; and (iii) estimation and inference about these different target parameters. Our approach allows for estimation and inference on interpretable causal parameters allowing for arbitrary treatment effect heterogeneity and dynamic effects, thereby completely avoiding the issues of interpreting results of standard two-way fixed effects (TWFE) regressions as causal effects in DiD setups as pointed out by [Borusyak and Jaravel \(2017\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Sun and Abraham \(2020\)](#), and [Athey and Imbens \(2018\)](#). In addition, it adds transparency and objectivity to the analysis ([Rubin \(2007, 2008\)](#)), and allows researchers to exploit a variety of estimation methods to answer different questions of interest.

The identification step of the analysis provides a blueprint for the other steps. In this paper, we pay particular attention to the disaggregated causal parameter that we call the *group-time average treatment effect*, i.e., the average treatment effect for group g at time t , where a “group” is defined by the time period when units are first treated. In the canonical DiD setup with two periods and two groups, these parameters reduce to the ATT which is typically the parameter of interest in that setup. An attractive feature of the group-time average treatment effect parameters is that they do not directly restrict heterogeneity with respect to observed covariates, the period in which units are first treated, or the evolution of treatment effects over time. As a consequence, these easy-to-interpret causal parameters can be directly used for learning about treatment effect heterogeneity, and/or to construct many other more aggregated causal

¹See Section 6 of [Athey and Imbens \(2006\)](#) and Theorem S1 in [de Chaisemartin and D’Haultfœuille \(2017\)](#) for notable exceptions that cover multiple periods and multiple groups.

that are potentially associated with the outcome evolution of $Y(0)$ during post-treatment periods. We explicitly rule out incorporating post-treatment covariates as they can potentially be affected by the treatment; see, e.g., [Wooldridge \(2005b\)](#), for a related discussion under the unconfoundedness setup.

3 Summarizing Group-Time Average Treatment Effects

The previous section shows that we can identify the $ATT(g, t)$'s by restricting treatment anticipation behavior and imposing a conditional parallel trends assumption. In many applications, the $ATT(g, t)$'s can be the ultimate causal parameters of interest. They can be used to highlight treatment effect heterogeneity across different groups g , at different points in time t , and across different lengths of treatment exposure, $e = t - g$. In other situations, however, researchers may want to combine these different $ATT(g, t)$'s to form more aggregated causal parameters. For instance, if the number of groups and time periods is relatively large, it may be challenging to interpret many group-time average treatment effects.

In this section, we consider different aggregation schemes for the $ATT(g, t)$'s that allow researchers to form a variety of summary measures of the causal effects of a given policy. Our aggregation schemes are of the form

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\tau} w(g, t) \cdot ATT(g, t), \quad (3.1)$$

where $w(g, t)$ are carefully-chosen (known or estimable) weighting functions specified by the researcher such that θ can be used to address a well-posed empirical/policy question. Different choices of $w(g, t)$ allows researchers to highlight different types of treatment effect heterogeneity. We pay particular attention to aggregations that result in a single overall treatment effect summary parameter as well as to aggregations related to understanding dynamic effects as is commonly done in event-study analysis. Of course, many other aggregated parameters of the type (3.1) can be easily constructed following our framework. We illustrate this point by also summarizing heterogeneity with respect to group or by calendar time.

Before proceeding with the discussion on how to construct these different aggregated parameters, it is worth revisiting the two most popular treatment effect summary measures used by practitioners in DiD setups. These are based on the “static” and “dynamic” two-way fixed effects (TWFE) linear regression specifications

$$Y_{i,t} = \alpha_t + \alpha_g + \beta D_{i,t} + \epsilon_{i,t}, \quad (3.2)$$

$$Y_{i,t} = \alpha_t + \alpha_g + \sum_{e=-K}^{-2} \delta_e^{anticip} \cdot D_{i,t}^e + \sum_{e=0}^L \beta_e \cdot D_{i,t}^e + v_{i,t}, \quad (3.3)$$

respectively, where α_t is a time fixed effect, α_g is a group fixed effect, $\epsilon_{i,t}$ and $v_{i,t}$ are error terms, $D_{i,t}^e = 1\{t - G_i = e\}$ is an indicator for unit i being e periods away from initial treatment at time t , and K and L are positive constants. The parameter of interest in the static TWFE specification is β , which, in applications, is typically interpreted as an overall effect of participating in the treatment across groups and time periods. In the dynamic TWFE specification, practitioners usually focus on the β_e , $e \geq 0$, and these parameters are typically interpreted as measuring the effect of participating in the

treatment at different lengths of exposure to the treatment.

Despite the popularity of these specifications, recent research has shown that one must be very careful in attaching a causal interpretation to these aggregated parameters. For instance, [Borusyak and Jaravel \(2017\)](#), [Goodman-Bacon \(2019\)](#), [de Chaisemartin and D’Haultfoeuille \(2020\)](#), and [Athey and Imbens \(2018\)](#) have shown that, in general, β recovers a weighted average of some underlying treatment effect parameters but some of the weights on these parameters can be negative. This can potentially lead to particularly problematic cases such as the effect of the treatment being positive for all units, but the TWFE estimation resulting in estimates of β that are negative. Even in cases where the weights are not negative, the weights on underlying treatment effect parameters are entirely driven by the TWFE estimation strategy and are sensitive to the size of each group, the timing of treatment, and the total number of time periods (see Theorem 1 in [Goodman-Bacon \(2019\)](#)). The results in this section can be used in exactly the same setup to identify a single interpretable average treatment effect parameter and, thus, provide a way to circumvent the issues with the more common approach.

As discussed by [Goodman-Bacon \(2019\)](#), the “negative weight problem” associated with β arises when treatment effects evolve over time. Thus, one may wonder if such problems would still be present when considering more general, dynamic specifications such as (3.3). [Sun and Abraham \(2020\)](#) shows that this is still the case as the β'_e s associated with (3.3) do not recover easy-to-interpret causal parameters and still generally suffer from the same sorts of “negative weighting problems.” In contrast to this, we provide a simple way to directly aggregate our group-time average treatment effects into average treatment effects across different lengths of exposure to the treatment.

3.1 Aggregations to Highlight Treatment Effect Heterogeneity

Next, we discuss several partial aggregations of the group-time average treatment effects in order to summarize different dimensions of treatment effect heterogeneity. Although there are additional possibilities, we focus our discussion below on how to answer three particular questions: (a) How does the effect of participating in the treatment vary with length of exposure to the treatment? (b) Do groups that are treated earlier have, on average, higher/lower average treatment effects than groups that are treated later? (c) What is the cumulative average treatment effect of the policy across all groups until some particular point in time? Throughout this section, to avoid notation clutter, we assume that units do not anticipate treatment, i.e., we consider the case where Assumption 3 holds with $\delta = 0$. We also assume that a “never treated” group is available.

3.1.1 How do average treatment effects vary with length of exposure to the treatment?

One of the most popular questions that arises in DiD setups with multiple time periods concerns treatment effect dynamics: How does the effect of participating in the treatment vary with length of exposure to the treatment? For instance, do average treatment effects increase/decrease with elapsed treatment time? Indeed, answering this type of question is often the main motivation for using the event study regression in (3.3), though, as we mentioned above, that sort of regression may not be suitable for such a task. In this section, we propose an aggregation scheme that is suitable to highlight treatment effect heterogeneity with respect to length of exposure to the treatment that does not suffer from the drawbacks associated

earnings gains induced by the training program. Alternatively, in health applications, researchers may want to measure how many COVID-19 cases have been averted by shelter-in-place orders up to day $\tilde{t}$. To consider the cumulative effect, consider the following parameter

$$\theta_c^{cumu}(\tilde{t}) = \sum_{t=2}^{\tilde{t}} \theta_c(t). \quad (3.9)$$

$\theta_c^{cumu}(\tilde{t})$ can be interpreted as the cumulative average treatment effect among the units that have been treated by time $\tilde{t}$.

3.2 Aggregations into Overall Treatment Effect Parameters

Finally in this section, we consider some ideas for aggregating group time average treatment effects into an overall effect of participating in the treatment. One very simple idea is to just average all of the identified group-time average treatment effects together; i.e., to consider the parameter

$$\theta_W^O = \frac{1}{\kappa} \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \mathbf{1}\{t \geq g\} ATT(g, t) P(G = g | G \leq \mathcal{T}) \quad (3.10)$$

where $\kappa = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\mathcal{T}} \mathbf{1}\{t \geq g\} P(G = g | G \leq \mathcal{T})$ (which ensures that the weights on $ATT(g, t)$ in the second term sum up to one). θ_W^O is a weighted average of each $ATT(g, t)$ putting more weight on $ATT(g, t)$'s with larger group sizes. Unlike β in the TWFE regression specification (3.2), this simple combination of $ATT(g, t)$'s immediately rules out troubling issues due to negative weights; as a particular example, when the effect of participating in the treatment is positive for all units, this aggregated parameter cannot be negative.

That being said, just requiring positive weights is a very minimal requirement of a reasonable overall treatment effect parameter. For example, one drawback of θ_W^O is that it systematically puts more weight on groups that participate in the treatment for longer. Instead, we suggest the following parameter as a general-purpose summary of the average effect of participating in the treatment

$$\theta_{sel}^O = \sum_{g \in \mathcal{G}} \theta_{sel}(g) P(G = g | G \leq \mathcal{T}) \quad (3.11)$$

where $\theta_{sel}(g)$ is the average effect of participating in the treatment for units in group g as defined in Equation (3.7) above. θ_{sel}^O first computes the average effect for each group (across all time periods) and then averages these effects together across groups to summarize the overall average effect of participating in the treatment. Thus, θ_{sel}^O is the average effect of participating in the treatment experienced by all units that ever participated in the treatment. In this respect, its interpretation is the same as the ATT in the canonical DiD setup with two periods and two groups. This is an attractive property for a summary measure of the overall effect of participating in the treatment in the context of multiple time periods and variation in treatment timing.

Working by analogy, one can also define overall treatment effect parameters by averaging $\theta_{es}(e)$ across